

## Advanced Heat Conduction

Dr. Debasis Sarkar

Department of Chemical Engineering

Indian Institute of Technology Kharagpur

### Reference Books

Heat Conduction - Ozisik

Heat Conduction – Yener and Kakac

| Sl No. | Day                         | Topic Covered                                                                                                                                                                      | Reading                                         |
|--------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| 1      | 8/1/2024                    | Introduction, Mechanisms of heat transfer, Fourier law, Vector form, Thermal conductivity tensor, Divergence theorem, Heat conduction equation, Moving solids, Boundary conditions | Ozisik – Ch 1;<br>YK – Ch1, Ch2;<br>Class notes |
| 2      | 15/1/2024                   | Nonlinear heat conduction equation, Kirchhoff transformation, Numerical example, Non-Fourier heat conduction, Single-phase lagging model, Dual-phase lagging model, Biot number    | Ozisik – Ch 1;<br>YK – Ch1, Ch2;<br>Class notes |
| 3      | 25/1/2024<br>(1 hour class) | Non-dimensional heat equation, Fourier number, Example, Separation of variables, Uniqueness of solution of heat equation                                                           | Class notes                                     |
| 4      | 29/1/2024                   | Fins - uniform and non-uniform cross-section, Bessel equations, Example, Fin optimization                                                                                          | Ozisik – Ch 1;<br>YK – Ch3;<br>Class notes      |
|        |                             | To be continued                                                                                                                                                                    |                                                 |
|        |                             |                                                                                                                                                                                    |                                                 |
|        |                             |                                                                                                                                                                                    |                                                 |